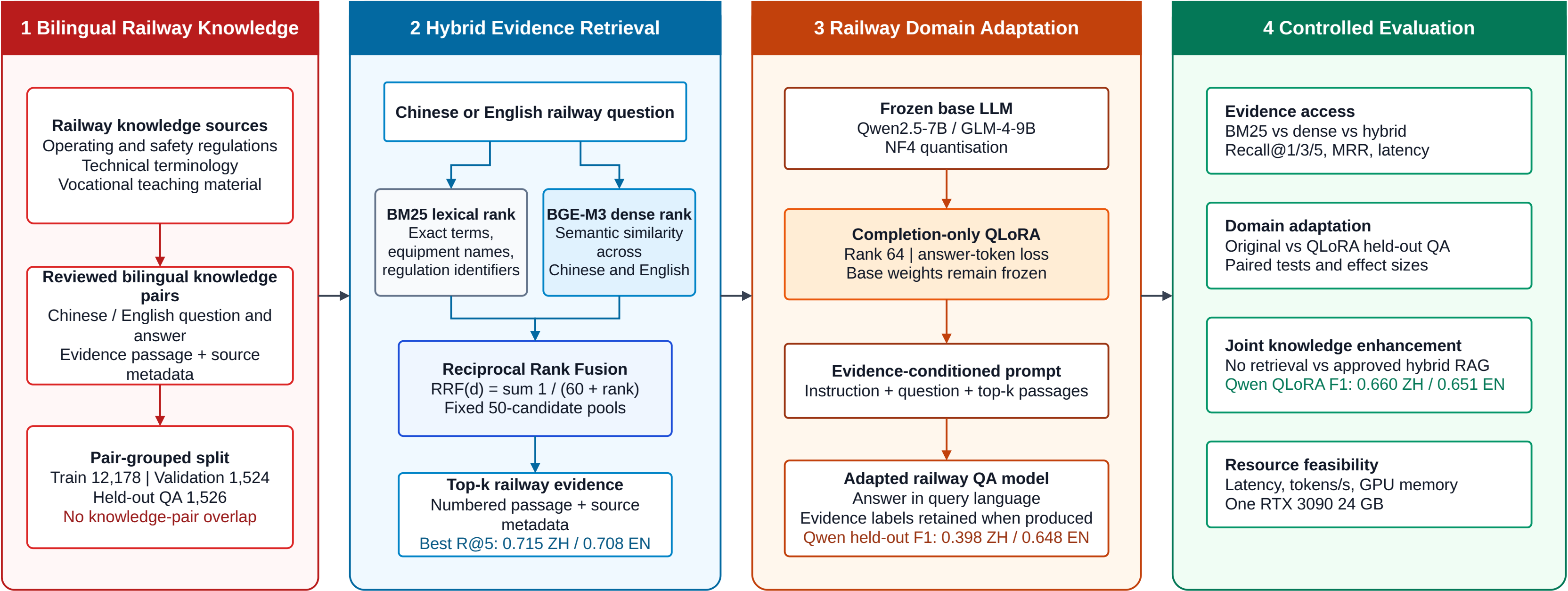


Knowledge-Enhanced Adaptation of a Railway Domain LLM

Hybrid evidence retrieval and parameter-efficient domain adaptation under a single-workstation constraint



Complementary gains: hybrid retrieval adds railway evidence; QLoRA specialises answer behaviour; both remain measurable within the local resource boundary.